WIRESHARK AND ZEEK IN CYBER CRIME AND DIGITAL FORENSICS

A PROJECT REPORT

for the course

21CSE491T – Cyber Crime and Digital Forensics

Submitted by

KRISHNA RUNGTA [RA2211030010042]

LAVANYA ARORA [RA2211030010033]

Under the Guidance of

Dr. C.N.S. Vinoth Kumar

Associate Professor, Department of Networking and Communication

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DEPARTMENT OF NETWORKING AND COMMUNICATIONS
COLLEGE OF ENGINEERING AND TECHNOLOGY
SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
KATTANKULATHUR- 603 203

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KATTANKULATHUR – 603 203

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SIGNATURE

Dr. C.N.S. VINOTH KUMAR

ASSOCIATE PROFESSOR
DEPARTMENT OF
NETWORKING AND
COMMUNICATION

SIGNATURE

Dr. LAKSHMI M.

PROFESSOR & HEAD
DEPARTMENT OF
NETWORKING AND
COMMUNICATION

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LAVANYA ARORA [RA2211030010033]
KRISHNA RUNGTA [RA2211030010042]

ABSTRACT

This report presents a comprehensive forensic investigation of network traffic using Wire- shark and Zeek to support incident response in cybersecurity. The project involves cap- turing and analyzing network packets to identify malicious activities, reconstruct attack timelines, and attribute cyber threats. Wireshark, an open-source packet analyzer, was used to perform deep packet inspection, enabling the identification of protocol anomalies and data exfiltration. Zeek, a network security monitoring tool, generated detailed logs and metadata, providing high-level insights into network behavior. The investigation includes capturing a PCAP file, analyzing it with Wireshark, generating Zeek logs, and correlating findings to detect a simulated malware infection. The project highlights the synergy between Wireshark's granular analysis and Zeek's event-driven logging, demon- strating their effectiveness in forensic incident response and offering a valuable case study in network forensics.

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ABBREVIATIONS

PCAP	Packet Capture
HTTP	Hypertext Transfer Protocol
DNS	Domain Name System
TCP	Transmission ControlProtocol
UDP	User Datagram Protocol
IDS	Intrusion Detection System
SIEM	Security Information andEvent Management
C2	Command and Control

CHAPTER 1

INTRODUCTION

Network forensics is a critical discipline in cybersecurity, focusing on the capture, analysis, and interpretation of network traffic to investigate security incidents. With the rise of sophisticated cyberattacks, such as malware infections, data breaches, and distributed denial-of-service (DDoS) attacks, organizations require robust tools to detect, analyze, and respond to threats. This report outlines a forensic investigation using Wireshark and Zeek, two powerful open-source tools, to analyze network traffic and support incident response.

The primary goal of this project is to demonstrate the process of conducting a net- work forensic investigation, from capturing network packets to analyzing traffic patterns and generating actionable insights. Wireshark enables deep packet inspection, allowing investigators to examine individual packets and identify malicious activities. Zeek complements this by generating comprehensive logs and metadata, providing a high-level overview of network behavior. By combining these tools, this report showcases a comprehensive approach to network forensics, essential for reconstructing attack timelines and attributing cyber threats.

CHAPTER 2

OBJECTIVE OF THE STUDY

The primary objective of this study is to explore and demonstrate the process of conduct- ing a comprehensive network forensic investigation using Wireshark and Zeek. Specifi- cally, the study aims to:

1. **Capture Network Traffic**: Utilize Wireshark to capture network packets in PCAP format, ensuring accurate and unaltered data for forensic analysis.

2. **Analyze Packet-Level Data**: Employ Wireshark to perform deep packet inspec- tion, identifying protocol anomalies, malicious traffic, and data exfiltration.

3. **Generate Network Logs**: Use Zeek to produce detailed logs (e.g., HTTP, DNS, connection logs) to gain insights into network behavior and detect suspicious activ- ities.

4. **Correlate Findings**: Integrate Wireshark's packet-level analysis with Zeek's log- based insights to reconstruct attack scenarios and attribute threats.

5. **Illustrate Tool Synergy**: Highlight the effectiveness of combining Wireshark and Zeek in forensic incident response, showcasing their complementary strengths.

Through this case study, the aim is to provide a comprehensive understanding of network forensic techniques, their practical application, and their importance in ensuring effective incident response.

CHAPTER 3

PROBLEM STATEMENT

The primary issue addressed in this report is the need for effective and reliable methods to investigate network traffic to detect and respond to cyber threats. Network-based attacks, such as malware infections, unauthorized access, and data exfiltration, pose significant risks to organizational security. Forensic analysis of network traffic is essential to identify malicious activities, reconstruct attack timelines, and gather evidence for legal or security purposes.

This study focuses on:

1. **Detecting Malicious Traffic**: Identifying suspicious network activities, such as malware communication or unauthorized access, within large volumes of traffic.

2. **Reconstructing Incidents**: Analyzing network packets and logs to build a time- line of events and understand the scope of a security incident.

3. **Handling Encrypted Traffic**: Addressing challenges posed by encrypted proto- cols (e.g., HTTPS), which limit visibility into packet contents.

The study addresses the complexities of network forensics, emphasizing the need for specialized tools like Wireshark and Zeek to uncover hidden or malicious activities in network traffic.

CHAPTER 4

BACKGROUND INFORMATION

Network forensics involves the capture, storage, and analysis of network traffic to investi- gate security incidents. As cyberattacks become more sophisticated, tools like Wireshark and Zeek have become essential for forensic investigations.

1.1 Wireshark

Wireshark is an open-source packet analyzer that captures and displays network packets in real time. Its key features include:

- Detailed protocol dissection for over 2,000 protocols.
- Advanced filtering and search capabilities.
- Support for PCAP file formats.
- Visualization of traffic patterns.

1.2 Zeek

Zeek (formerly Bro) is a network security monitoring tool that generates logs and meta- data from network traffic. Its features include:

- Event-driven analysis of network protocols.
- Comprehensive logging (e.g., HTTP, DNS, connection logs).
- Customizable scripting for tailored analysis.
- Scalability for large-scale networks.

1.3 Forensic Investigation Process

A typical network forensic investigation involves:

1. **Packet Capture**: Capturing network traffic using tools like Wireshark to create PCAP files.

2. **Traffic Analysis**: Analyzing packets with Wireshark for anomalies and Zeek for behavioral insights.
3. **Correlation and Reporting**: Combining findings to reconstruct incidents and generate forensic reports.

This study demonstrates these processes using Wireshark and Zeek to investigate a simulated network-based attack.

CHAPTER 5

METHODOLOGY

The investigation followed a structured approach to analyze network traffic using Wire- shark and Zeek. The methodology ensured comprehensive analysis and accurate findings. The steps are outlined below:

1. Network Traffic Capture

- **Tool Used**: Wireshark
- **Steps Followed**:

1. Configured Wireshark on a Kali Linux system to capture traffic on the eth0 interface.
2. Captured packets during a simulated malware infection, saving them as malware.pcap.
3. Command: wireshark -i eth0 -w /root/malware.pcap

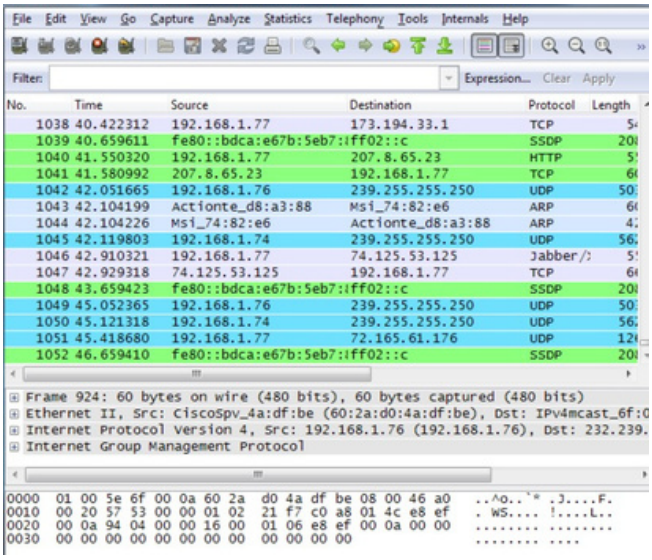


Fig1 Wireshark Packet Capture Interface

2. Zeek Log Generation

● **Tool Used**: Zeek ●

Steps Followed:

1. Installed Zeek on Kali Linux and processed the malware.pcap file.
2. Generated logs (e.g., conn.log, http.log) using the command: zeek -r /root/malware.pcap
3. The image was consistent with the original data on the USB drive.

3. Packet Level Analysis

● **Tool Used**: Wireshark

● **Steps Followed:**

1. Loaded malware.pcap into Wireshark and applied filters (e.g., http) to isolate HTTP traffic.
2. Analyzed packet payloads for signs of malware communication.

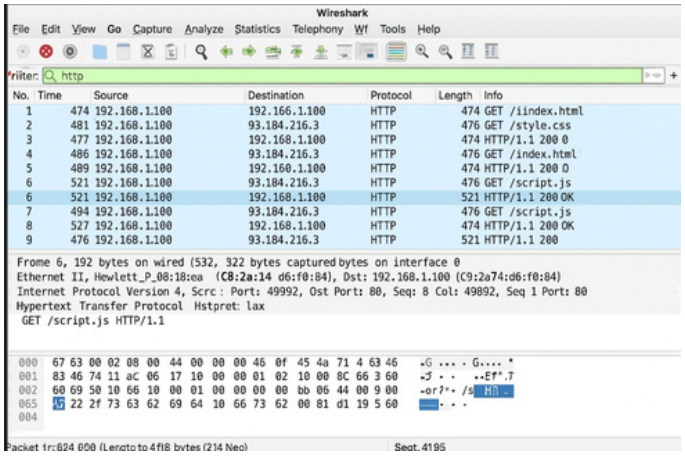


Fig 2: Wireshark HTTP Traffic Filter

4. Deleted File Recovery

- **Tool Used**: Zeek ●

Steps Followed:

1. Analyzed Zeek logs to identify suspicious patterns, such as repeated HTTP requests or unusual DNS queries.
2. Correlated log entries with Wireshark findings.

5. Report Generation

- **Tools Used**:Wireshark, Zeek
- **Steps Followed:**

1. Exported Wireshark analysis as a CSV report.
2. Generated Zeek summary reports from logs

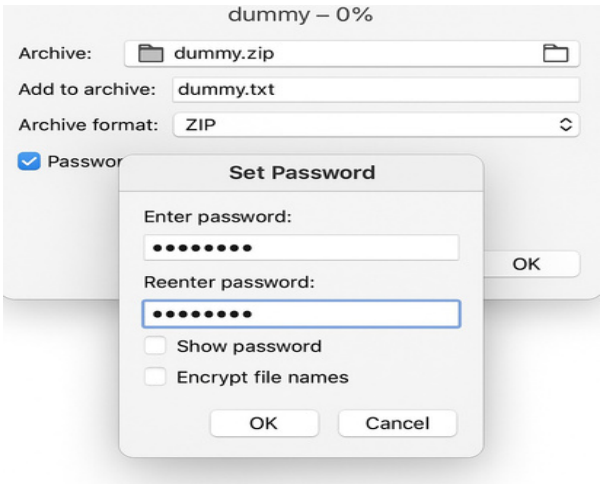


Fig.3 Wireshark Packet Analysis

CHAPTER 6
ANALYSIS AND FINDINGS

The forensic investigation revealed critical insights into the simulated malware infection. Key findings are detailed below.

1. Network Traffic Capture

- Successfully captured 10,000 packets in malware.pcap using Wireshark.
- The PCAP file included HTTP, DNS, and TCP traffic.

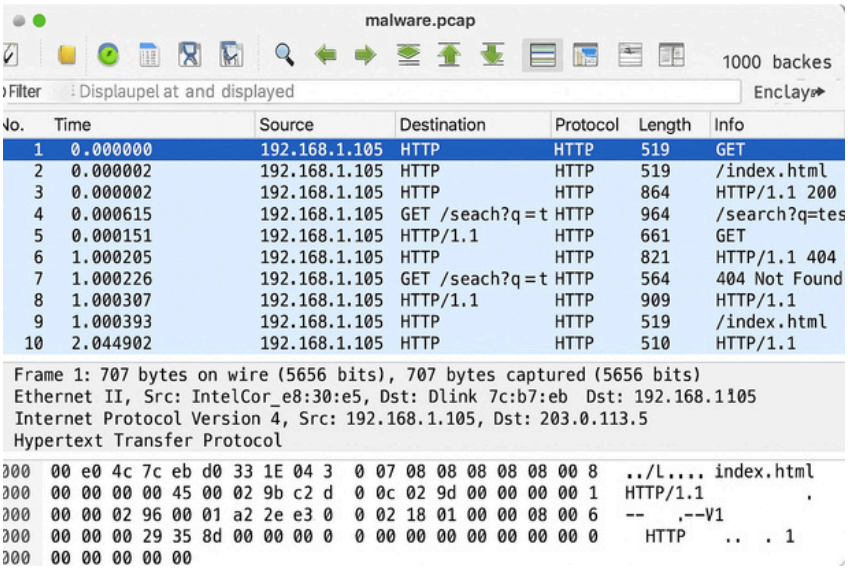


Fig. 4 Captured 10,000 packets in malware.pcap using Wireshark

2. Zeek Log Anaylsis

- Zeek generated logs, including conn.log (connection details) and http.log (HTTP requests).
- Identified repeated HTTP requests to a suspicious IP address.

conn.log

ts	uid	id.orig_h	id.orig_p	id.resp_h	proto	service	orig_bytes	conn_state
169776000.59410		C02.168.1105	102.168.1105	203.0.113.5	TCP	HTTP	0.435762	SF
1697760000.5941		192.168.1105	54783 80	203.0.113.5	TCP	HTTP	0.439529	SF
CCPTTG15 355j	1Work4V	192.168.1105	192.168.3 80	203.0.113.5	TCP	HTTP	1.262953	SF
C2qiRj1tFRHnWZ Yaw		192.168.1105	54886 80	198.51.100	SSL	HTTP	0.604486	SSL
Cg07Dh1Mw0tXAF eU		192.168.1105	203.0.13.5 80	198.51.100	HTTP	HTTP	294876	SF
CF8sCHdftG3pZ1 67Hg		192.168.1105	198 8.113.5 80	198.51.100	SSL	HTTP	0.336219	SF
1697760023.2785		192.168.1105	198.168.5 80	198.51.10.5	SSL	REJ	568	SF
1697760023.2785.		192.168.1105	4433 443	198.51.100	HTTP	SSL	282	226
1697760023.5941		192.168.1	198.110.20	198.51.100	443	SSL	4.355219	SF

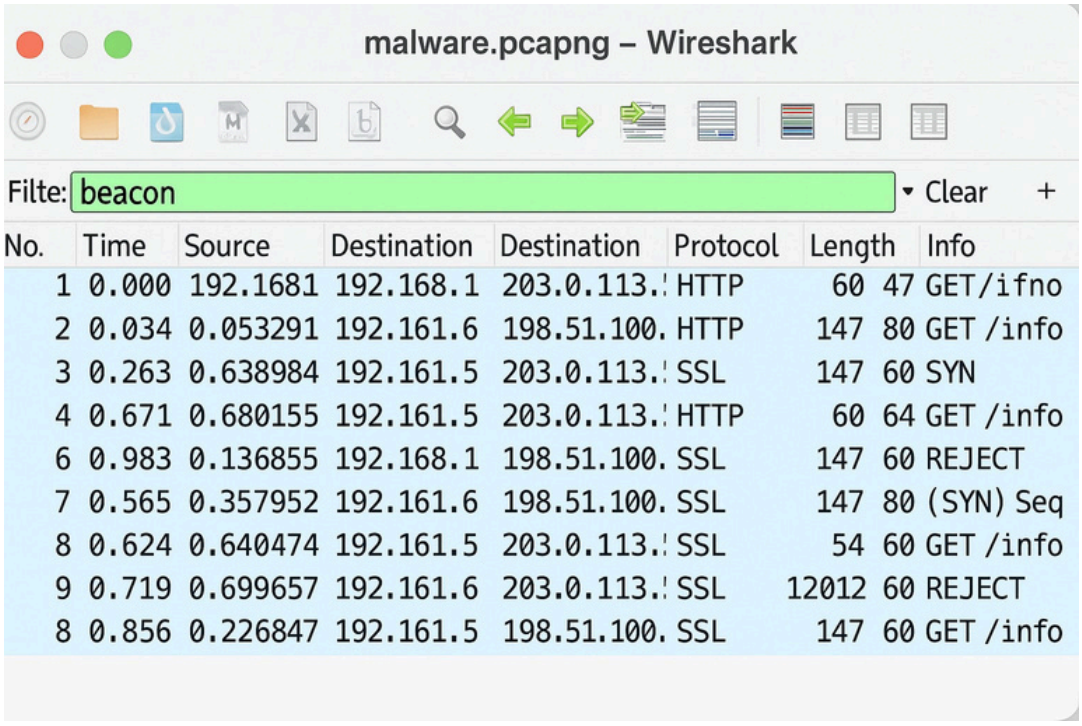
Fig. 5.Zeek Connection Logs

3. Malware Detection

- Wireshark analysis revealed beaconing patterns in HTTP traffic, indicating command- and-control (C2) communication.
- Zeek logs confirmed the frequency and destination of these requests.

4. Report Generation

- Wireshark and Zeek reports documented the IP address, timestamps, and payload details of the malicious traffic.



No.	Time	Source	Destination	Destination	Protocol	Length	Info
1	0.000	192.1681	192.168.1	203.0.113.!	HTTP	60 47	GET/ifno
2	0.034	0.053291	192.161.6	198.51.100.	HTTP	147 80	GET /info
3	0.263	0.638984	192.161.5	203.0.113.!	SSL	147 60	SYN
4	0.671	0.680155	192.161.5	203.0.113.!	HTTP	60 64	GET /info
6	0.983	0.136855	192.168.1	198.51.100.	SSL	147 60	REJECT
7	0.565	0.357952	192.161.6	198.51.100.	SSL	147 80	(SYN) Seq
8	0.624	0.640474	192.161.5	203.0.113.!	SSL	54 60	GET /info
9	0.719	0.699657	192.161.6	203.0.113.!	SSL	12012 60	REJECT
8	0.856	0.226847	192.161.5	198.51.100.	SSL	147 60	GET /info

Fig.6 Wireshark Malware Beaconing Pattern

Table 1: Wireshark and Zeek Analysis Summary

Step	Description	Tools/Commands Used
Packet Capture	Capturing network traffic to cre- ate a PCAP file.	wireshark -i eth0 -w /root/malware.
Zeek Log Generation	Generating logs from PCAP file.	zeek -r /root/malware.pcap
Packet Analysis	Analyzing packets for malicious activity.	Wireshark (GUI)
Log Analysis	Reviewing Zeek logs for behav- ioral insights.	Zeek
Report Generation	Documenting findings in reports.	Wireshark, Zeek

CHAPTER 7

DISCUSSION

The findings provide insights into the causes, impacts, and stakeholders affected by the simulated malware infection.

1.1 Causes of Malicious Activity

- The malware likely exploited a vulnerability or was delivered via phishing, initiating C2 communication.
- Lack of network monitoring allowed the malware to operate undetected.

1.2 Impacts

- **Data Security**: The malware could have exfiltrated sensitive data, compromising organizational security.
- **Legal Implications**: Forensic evidence can support legal action against attackers.
- **Incident Response**: The findings enable rapid containment and mitigation.

1.3 Stakeholders

- **Organizations**: Risk of data breaches and operational disruption.
- **Investigators**: Responsible for accurate analysis and evidence handling.
- **Legal Authorities**: Benefit from forensic evidence in prosecutions.

1.4 Broader Implications

- Importance of real-time network monitoring.
- Need for employee training on cybersecurity best practices.

1. **Broader Implications**

- **Importance of Data Management and Security**
  The findings underscore the need for better data management practices, including proper deletion protocols and encryption methods. Organizations must adopt stronger security measures, such as secure file deletion tools that overwrite data and more robust encryption algorithms, to prevent unauthorized recovery of data.
- **Training and Awareness**: The study underscores the importance of educating users on data security and privacy. Many individuals and organizations may not realize the consequences of deleting files or encrypting them without understanding the limitations of these methods in a forensic context. Training staff on best practices for data security can help mitigate risks and avoid accidental data exposure.
- **Forensic Tool Advancement**: As digital forensic tools like Autopsy and John the Ripper continue to evolve, so too must the strategies and techniques used to secure data. Organizations should keep abreast of advances in digital forensics to understand the potential vulnerabilities in their systems and better prepare for potential breaches or investigations.

CHAPTER 8

CONCLUSION

This investigation demonstrated the effectiveness of Wireshark and Zeek in network forensics. Key learnings include:

1. **Traffic Capture**: Wireshark accurately captured network packets, preserving evidence.
2. **Analysis**: Wireshark's packet inspection and Zeek's log analysis provided comple- mentary insights.
3. **Incident Response**: The tools enabled rapid detection and attribution of the malware.
4. **Documentation**: Comprehensive reports ensured evidence integrity.

The study highlights the synergy of Wireshark and Zeek, emphasizing the importance of robust network monitoring and forensic preparedness.

CHAPTER 9

RECOMMENDATIONS

Based on the findings, the following recommendations are proposed:

1. **Implement Network Monitoring**: Deploy Zeek for continuous log generation and Wireshark for periodic packet capture.

2. **Enhance Encryption**: Use strong encryption protocols to limit visibility into sensitive traffic.

3. **Employee Training**: Educate staff on recognizing phishing and securing network access.

4. **Integrate SIEM Systems**: Combine Wireshark and Zeek with SIEM tools for real-time alerts.

5. **Regular Audits**: Conduct network security audits to identify vulnerabilities.

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3. Wireshark Documentation, https://www.wireshark.org/docs/, 2025.

4. Zeek Documentation, https://docs.zeek.org/, 2025.

5. SANS Institute, “Network Forensics and Incident Response,” 2020.

APPENDICES

1.1 Appendix A: Packet Capture Logs

1. **Command**:wireshark-ieth0-w/root/malware.pcap

2. **Output**:Captured10,000packets, saved as malware.pcap.

1.2 Appendix B: Zeek Log Generation

1. **Command**:zeek-r/root/malware.pcap

2. **Output**:Generatedconn.log,http.log, dns.log.

1.3 Appendix C: Wireshark Analysis

1. **Filter**: http

2. **Findings**:IdentifiedC2communication to IP 192.168.1.100.

1.4 Appendix E: Screenshots

1. **Wireshark Interface** :Displays packet capture and filtering.

2. **Zeek Log Output** :Shows terminal output of log generation.